

Project Title

Student Performance Tracker Using Machine Learning

1 Introduction

Student Performance Tracker is a Machine Learning project used to analyze and predict student academic performance based on attendance, study hours, assignment marks, and exam scores.

2 Objective

The main objective of this project is to track and predict student performance using Machine Learning techniques.

3 Problem Statement

Educational institutions need an efficient system to monitor student performance. Manual analysis takes time and may not be accurate. This project helps automate student performance analysis.

4 Technologies Used

- Python
- Jupyter Notebook / VS Code
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

5 Dataset Description

The dataset contains student details such as attendance percentage, study hours, assignment scores, exam scores, and performance status.

6 Methodology

1. Data Collection
2. Data Preprocessing
3. Data Visualization
4. Model Training

5. Prediction

6. Performance Analysis

7. Algorithm Used

Decision Tree Classifier

8. Advantages

- Easy performance tracking
- Reduces manual work
- Provides quick analysis
- Improves academic monitoring

9. Applications

- Schools
- Colleges
- Online Learning Platforms
- Educational Analytics Systems

10. Conclusion

This project helps educational institutions monitor and predict student performance efficiently using Machine Learning techniques.

11. Future Scope

Future improvements include adding web applications, dashboards, and advanced Deep Learning models for better predictions.